

The Geometry of Vision: The Horizon as Archimedean Place

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Abstract

Vision requires projective geometry. The eye projects a three-dimensional world onto a two-dimensional retina: a map between manifolds, a differential geometric operation performed by every seeing organism without knowledge of the mathematics. Parallel lines meet at the horizon. The horizon is the point at infinity — the archimedean place, the missing fiber of $\text{Spec } \mathbb{Z}$, the point the Arakelov compactification adds but cannot animate with a Frobenius. What you see when you look at the horizon is the archimedean place made visible. The vanishing point is the fundamental class [2] of the visual field: the point at which all lines balance, the fulcrum of perspective, the still lever [1] of sight. Vision is the geometry that $\text{Spec } \mathbb{Z}$ is missing — the projective completion, the containing fiber, seen directly by any eye that looks toward the horizon.

1 The Eye as Map

Definition 1 (Vision as projection). Vision is the map

$$\pi : \mathbb{R}^3 \setminus \{O\} \rightarrow S^2$$

from three-dimensional space (minus the eye O) to the unit sphere (the retina), given by central projection: $\pi(P) = (P - O)/|P - O|$. Every point in the visual field is a ray from the eye. Two points on the same ray are indistinguishable to the eye: they project to the same retinal location. Vision does not see distance — it sees direction.

The geometry of vision is therefore the geometry of rays: *projective geometry*, in which a point is a line through the origin and the fundamental objects are incidence relations, not distances.

Proposition 2 (Parallel lines meet). *In Euclidean geometry, parallel lines do not meet. In the projective completion $\mathbb{RP}^2$ of $\mathbb{R}^2$, every family of parallel lines meets at a unique point at infinity — a direction, a limit, the place toward which the lines tend without ever arriving in $\mathbb{R}^2$.*

The horizon is the set of all points at infinity in the visual field: the projective line $\mathbb{RP}^1$ at the boundary of $\mathbb{RP}^2$. It is not a place you can walk to. It is the containing boundary of

the visual space — the point the eye adds to complete the geometry. Without the horizon, the visual field is open, non-compact, without a fundamental class. With the horizon, the visual field is projectively complete: contained, with a fundamental class, with a fulcrum.

2 Brunelleschi and the Vanishing Point

Proposition 3 (The discovery of projective geometry). *In 1415, Filippo Brunelleschi stood at the door of the Florence Cathedral and held a painted panel of the Baptistery with a small hole at the vanishing point. He looked through the hole from behind at a mirror held at arm's length, reflecting the painting back. The painted perspective and the actual building coincided exactly.*

He had discovered the vanishing point: the single point at infinity toward which all parallel horizontal lines converge. The vanishing point is the fundamental class of the visual field — the point from which the entire perspective is measured, the fulcrum at which all lines balance, the still lever [1] of sight made visible.

Brunelleschi was doing algebraic geometry without knowing it. The vanishing point is a point of $\mathbb{RP}^2 \setminus \mathbb{R}^2$: a point at infinity, real and visible, the archimedean place of perspective.

3 The Horizon as Archimedean Place

Theorem 4 (The horizon is the missing fiber). *$\text{Spec } \mathbb{Z}$ has one point for each prime p : the fiber at p is $\text{Spec } \mathbb{F}_p$. It is missing the archimedean place: the fiber at ∞ , corresponding to $\mathbb{R}$ (or $\mathbb{C}$). The Arakelov compactification $\overline{\text{Spec } \mathbb{Z}}$ adds this fiber, making the space complete.*

The analogy with vision:

<i>Arithmetic</i>	<i>Vision</i>
$\text{Spec } \mathbb{Z}$: primes $p = 2, 3, 5, \dots$	Visual space: points at finite distance
Missing archimedean fiber	Missing horizon
$\overline{\text{Spec } \mathbb{Z}}$: primes + ∞	$\mathbb{RP}^2$: finite points + line at infinity
Frobenius at p	Local geometry at each distance
Missing global Frobenius	No metric at the horizon
Arithmetic Wall	Vanishing point: visible but unreachable

The horizon is what $\overline{\text{Spec } \mathbb{Z}}$ is trying to be: the completing fiber, the point at infinity that closes the space, the archimedean place made present.

But the horizon is reachable by vision — you can see it — while the archimedean fiber of $\text{Spec } \mathbb{Z}$ cannot be animated: there is no Frobenius at ∞ , no eigenvalue, no purity. The eye completes its projective space with a horizon it can see. Arithmetic cannot complete $\text{Spec } \mathbb{Z}$ with a fiber it can use. The Wall is the gap between these two completions.

4 The Blind and the Sighted

Proposition 5 (Two geometries of space). *A sighted person's geometry is projective: the horizon exists, the vanishing point is present, parallel lines converge, the visual field is contained. The fundamental class of the visual field is present. The still lever holds: the horizon balances the near and the far.*

A person blind from birth has a different geometry: tactile, local, without a vanishing point. Space is constructed from proximity, texture, sound — all finite, all measurable, none projectively complete. The horizon does not exist in tactile geometry. The space is open, non-compact, without a point at infinity.

This is not a deficiency. It is a different grammar. The tactile geometry is the geometry of $\mathbb{R}^3$: rich, precise, locally Euclidean, globally non-compact. The visual geometry is the geometry of $\mathbb{RP}^3$: containing, projectively complete, with a fundamental class.

The Arithmetic Wall is the situation of $\text{Spec } \mathbb{Z}$: it has the local geometry (Frobenius at each prime p) but not the global completion (no Frobenius at ∞). It sees the finite distances but not the horizon. It is doing arithmetic in tactile geometry, pressing toward a visual completion it cannot achieve.

5 Perspective as Grammar

Theorem 6 (Perspective is cohomological grammar). *Perspective (Brunelleschi, Alberti, Dürer) is a cohomological grammar [3]: it reads what persists under projection. Persistent: incidence, cross-ratio, the horizon. Not persistent: distance, angle, area.*

The cohomology of the visual field is projective cohomology: $H^(\mathbb{RP}^2, \mathbb{Z}/2)$. The fundamental class is $[\mathbb{RP}^2] \in H^2(\mathbb{RP}^2, \mathbb{Z}/2)$: the class that the horizon bounds, the generator of the cohomology, the vanishing point generalized to all of $\mathbb{RP}^2$.*

Vision is the practice of finding what persists under projection. Mathematics [1] is the practice of finding what persists under every valid transformation. Vision is a special case of mathematics, performed by the eye before the proof arrives.

Remark 7 (The horizon and the proof). Every proof of the Riemann Hypothesis must deal with the archimedean place [4]: the L^2 estimate on the adèle class space that controls the behavior at ∞ . This is the horizon of the arithmetic visual field.

The proof would be the moment $\text{Spec } \mathbb{Z}$ learns to see its own horizon — to use the archimedean fiber not just as a boundary but as a fulcrum, a vanishing point, a fundamental class.

That moment: the arithmetic eye opens. The horizon is there. It was always there. The proof is the first clear sight of it.

References

- [1] Rincón, D. (2026). Mathematics. Preprint.
- [2] Rincón, D. (2026). The Fundamental Class: Is the Universe Contained? Preprint.

- [3] Rincón, D. (2026). Mathematics as Cohomological Grammar. Preprint.
- [4] Rincón, D. (2026). Week 10: Self-Adjointness. Preprint.
- [5] Rincón, D. (2026). The Arithmetic Wall. Preprint.